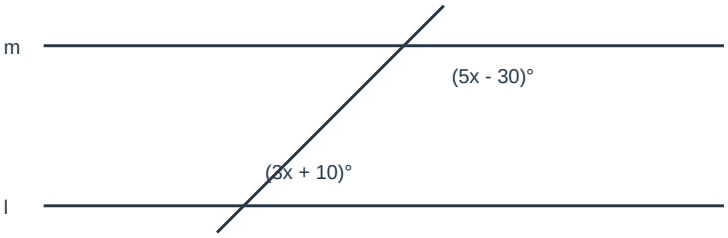
JambiX Mathematics

20 Challenging UTME-Style Diagram Questions

Mixed mathematics topics • Multiple-choice practice • Answers and explanations included

Instruction: Study each diagram and choose the correct option. Diagrams are illustrative and not necessarily drawn to scale.

1. Two parallel lines are cut by a transversal. Find x.



A. 15

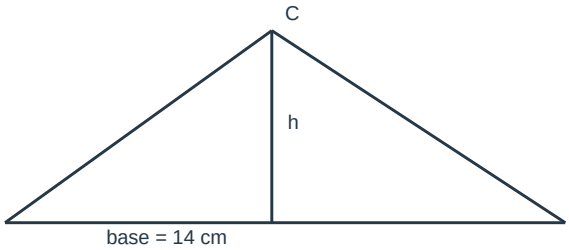
B. 20

C. 25

D. 30

Correct answer: B. Alternate interior angles are equal: 3x + 10 = 5x – 30. Hence 2x = 40 and x = 20.

2. The triangle has area 70 cm² and base 14 cm. Find its perpendicular height h.



A. 5 cm

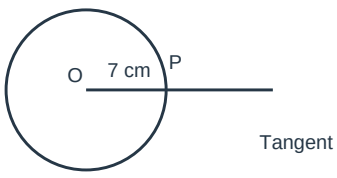
B. 8 cm

C. 10 cm

D. 12 cm

Correct answer: C. 70 = ½ × 14 × h = 7h. Therefore h = 10 cm.

3. OP is a radius meeting a tangent at P. What is the angle between OP and the tangent?



A. 45°

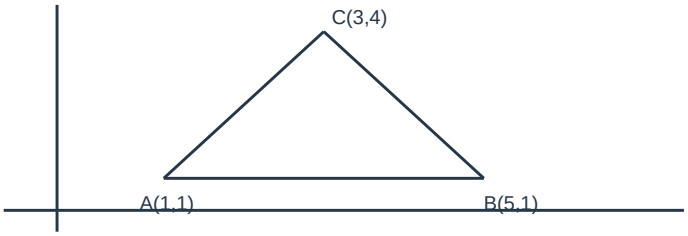
B. 60°

C. 90°

D. 120°

Correct answer: C. A radius is perpendicular to the tangent at the point of contact. The angle is 90°.

4. A triangle has vertices A(1,1), B(5,1) and C(3,4). Find its area.



A. 4

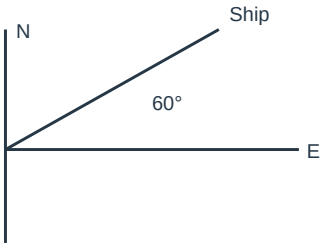
B. 6

C. 8

D. 10

Correct answer: B. AB = 4 units and the perpendicular height is 3 units. Area = ½ × 4 × 3 = 6 square units.

5. A ship travels 20 km at 60° north of east. Find its northward displacement.



A. 10 km

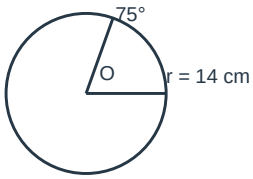
B. 10√3 km

C. 20√3 km

D. 20 km

Correct answer: B. Northward component = 20 sin 60° = 20 × √3/2 = 10√3 km.

6. Find the area of a sector of radius 14 cm and angle 75°. Use π = 22/7.



A. 110 cm²

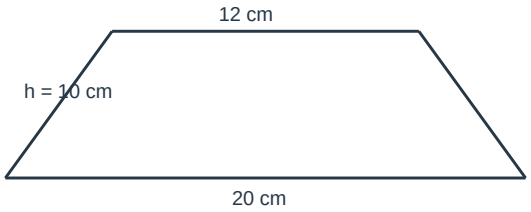
B. 128■ cm²

C. 140 cm²

D. 154 cm²

Correct answer: B. Area = 75/360 × π × 14² = 128■ cm².

7. The parallel sides of a trapezium are 12 cm and 20 cm. Its height is 10 cm. Find its area.



A. 120 cm²

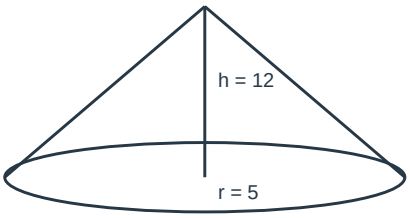
B. 140 cm²

C. 160 cm²

D. 180 cm²

Correct answer: C. Area = ½(12 + 20) × 10 = 160 cm².

8. A cone has radius 5 cm and vertical height 12 cm. Find its volume.



A. 60π cm³

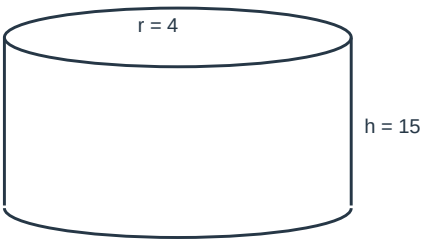
B. 75π cm³

C. 100π cm³

D. 150π cm³

Correct answer: C. Volume = ■πr²h = ■ × π × 25 × 12 = 100π cm³.

9. A cylinder has radius 4 cm and height 15 cm. Find its curved surface area.



A. 60π cm²

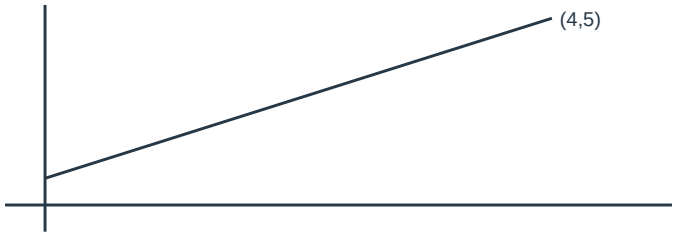
B. 100π cm²

C. 120π cm²

D. 160π cm²

Correct answer: C. Curved surface area = 2πrh = 2 × π × 4 × 15 = 120π cm².

10. A straight line passes through (0,1) and (4,5). Find its gradient.



A. 1

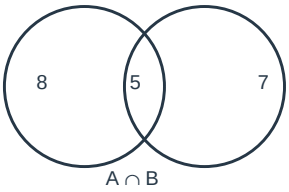
B. 2

C. 3

D. 4

Correct answer: A. Gradient = (5 – 1)/(4 – 0) = 1.

11. The Venn diagram gives 8 in A only, 5 in A ∩ B and 7 in B only. Find n(A ∪ B).



A. 13

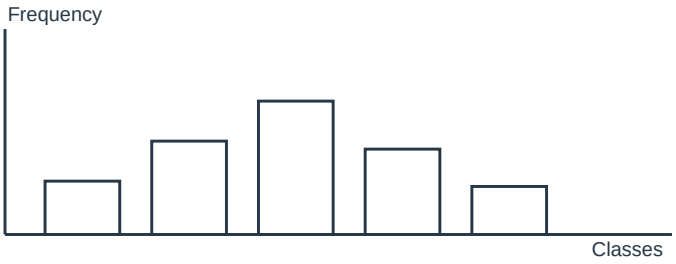
B. 15

C. 20

D. 25

Correct answer: C. n(A ∪ B) = 8 + 5 + 7 = 20.

12. A frequency chart has class frequencies 4, 6, 8, 5 and 3. Find the total frequency.



A. 21

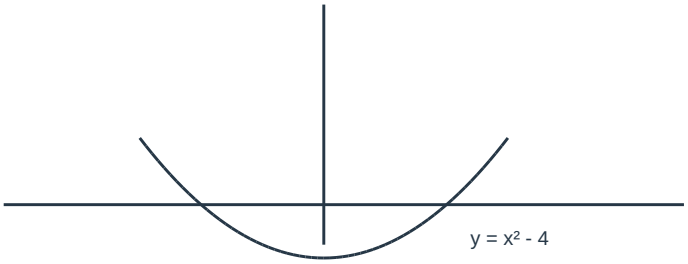
B. 24

C. 26

D. 28

Correct answer: C. Total frequency = 4 + 6 + 8 + 5 + 3 = 26.

13. The graph represents $y = x^2 - 4$. At which points does it cross the x-axis?



A. –4 and 4

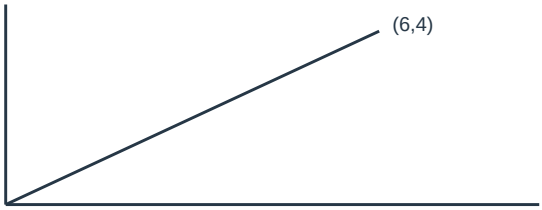
B. –2 and 2

C. 0 and 4

D. –1 and 1

Correct answer: B. At the x-axis, $y = 0$. Thus $x^2 - 4 = 0$, giving $x = -2$ or $x = 2$.

14. A vector has components 6 units east and 4 units north. Find its magnitude.



A. $2\sqrt{13}$

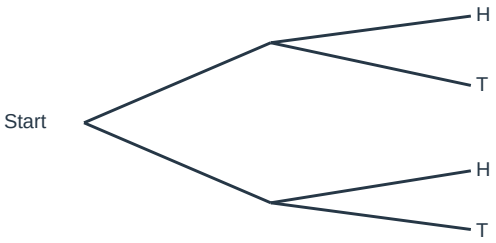
B. $2\sqrt{10}$

C. 10

D. 52

Correct answer: A. Magnitude = $\sqrt{(6^2 + 4^2)} = \sqrt{52} = 2\sqrt{13}$.

15. A fair coin is tossed twice. Find the probability of getting exactly one head.



A. ¼

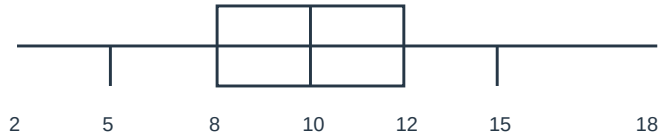
B. ■

C. ½

D. ¾

Correct answer: C. Possible outcomes are HH, HT, TH and TT. HT and TH contain exactly one head, so P = 2/4 = ½.

16. The box plot has Q■ = 8 and Q■ = 12. Find the interquartile range.



A. 2

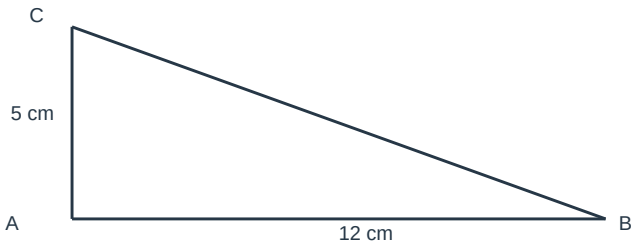
B. 4

C. 8

D. 12

Correct answer: B. Interquartile range = Q■ – Q■ = 12 – 8 = 4.

17. A right-angled triangle has perpendicular sides 5 cm and 12 cm. Find the hypotenuse.



A. 13 cm

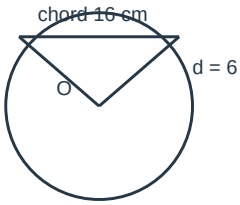
B. 15 cm

C. 17 cm

D. 19 cm

Correct answer: A. Hypotenuse² = 5² + 12² = 169. Therefore the hypotenuse is 13 cm.

18. A chord is 16 cm long and its perpendicular distance from the centre is 6 cm. Find the radius.



A. 8 cm

B. 9 cm

C. 10 cm

D. 12 cm

Correct answer: C. The perpendicular from the centre bisects the chord. Half-chord = 8 cm. Radius = √(8² + 6²) = 10 cm.

19. Given A = [[2,1],[3,4]] and B = [[1,2],[0,1]], find det(AB).

A = [2 1]
[3 4]

B = [1 2]
[0 1]

A. 5

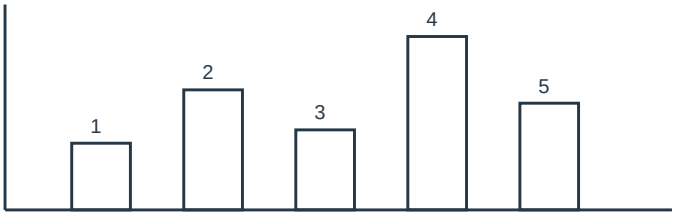
B. 7

C. 10

D. 12

Correct answer: A. det(AB) = det(A)det(B). det(A) = 8 – 3 = 5 and det(B) = 1. Thus det(AB) = 5.

20. A bar chart shows frequencies 25, 45, 30, 65 and 40. Find the total frequency.



A. 195

B. 200

C. 205

D. 210

Correct answer: C. Total frequency = 25 + 45 + 30 + 65 + 40 = 205.